The evaluation procedure for evaluating the limits

(1) Let $A = \lim_{x \to a} g(x)$. Then taking logarithm, with the assumption that $A > 0$ to ensure the continuity of the logarithm function, we get $\log A = \lim_{x \to a} \log(g(x))$. Therefore using the above theorem with $f(x) = \log x$ we have the limit

$$\lim_{x \to a} \log(g(x)) \;=\; \log\left(\lim_{x \to a} g(x)\right).$$

(2) We have the limit $\lim_{x \to a} \log(g(x))$ into either $\left(\frac{0}{0}\right)$ or $\left(\frac{\infty}{\infty}\right)$ form evaluate it using l'Hôpital Rule.

(3) Let that evaluated limit be say α. Then the required limit is e^{α}.

Example 7.43

Using the l'Hôpital Rule, prove that $\lim_{x \to 0^+} (1+x)^{\frac{1}{x}} = e$.

Solution

This is an indeterminate of the form 1^{∞}. Let $g(x) = (1+x)^{\frac{1}{x}}$. Taking the logarithm, we get

$$\log g(x) \;=\; \frac{\log(1+x)}{x}$$

$$\lim_{x \to 0^+} \log(g(x)) \;=\; \lim_{x \to 0^+} \left(\frac{\log(1+x)}{x}\right) \left(\frac{0}{0} \text{ form}\right)$$

$$= \lim_{x \to 0^+} \left(\frac{\frac{1}{1+x}}{1}\right) \qquad \text{(by l'Hôpital Rule)}$$

$$= 1 \,.$$

$$\text{But, } \lim_{x \to 0^+} \log g(x) \;=\; \log\left(\lim_{x \to 0^+} g(x)\right)$$

$$\text{Therefore, } \log\left(\lim_{x \to 0^+} g(x)\right) \;=\; 1.$$

Hence by exponentiating, we get $\lim_{x \to 0^+} g(x) = e$. ■

Example 7.44

Evaluate : $\lim_{x \to \infty} (1+2x)^{\frac{1}{2\log x}}$.

Solution

This is an indeterminate of the form ∞^0.

$$\text{Let } \; g(x) \;=\; (1+2x)^{\frac{1}{2\log x}} \,.$$

Taking the logarithm, we get

$$\log g(x) \;=\; \frac{\log(1+2x)}{2\log x}$$

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